MCO-1D Wing Manufacturing

Addendum R21 — TPU Winglet Section

Design rationale, geometry, manufacture, and commissioning of the TPU feather wingtip section for all four wing pairs.

Addendum	R21 — TPU Winglet Section
Date	August 2026
Applies to	Wing Manufacturing Document (original) + Addenda R20
Pairs	W1, W2, W3, W4 — all eight wings
Prerequisites	Wing Manufacturing Document complete through STEP FINAL-2. Rigid wing fully assembled, CF spar not yet cut to final length.
Commissioning	Two parameters must be measured and entered in software: wc_set_fuselage_width [mm] · wc_set_tpu_start [mm] Pre-flight check fails until both are confirmed.

R21-0 Purpose and Background

The MCO-1D wing has four structurally distinct elements: the leading-edge D-box (fibreglass over carbon fibre rod), PLA 3D-printed ribs, a carbon fibre trailing-edge rod, and a ripstop nylon membrane covering the aft portion. The inner span of this structure is rigid and carries aerodynamic bending loads. At high stroke amplitudes, the outer portion of each wing enters a zone where the port and starboard wings of the same pair approach each other.

At the stroke amplitudes required for clap-and-fling (above ~94°), the first contact between port and starboard wings occurs at approximately 74% span. This contact falls on the D-box and PLA ribs — fibreglass against fibreglass, PLA against PLA. These materials will chip, crack, and fatigue-fracture under repeated impact loads at flapping frequency. The wings would be destroyed.

The solution is to replace the outer 26–28% of wing span — the contact zone — with a single-piece TPU (thermoplastic polyurethane) section. TPU at Shore 85A tolerates thousands of contact cycles without damage. The contact becomes the intended aerodynamic event — the clap-and-fling — rather than a structural failure.

● **KEY POINT** The TPU section is not a compromise forced by manufacturing convenience. Research on slotted wingtips (Lynch et al. 2018; Dan et al. 2021) shows that four-fingered flexible wingtip devices at low Reynolds number improve lift coefficient by 7–20% compared to a solid tip. The feather section outperforms a solid tip in the MCO-1D's operating regime.

R21-1 Geometry and Dimensions

> ■ **CAUTION** All dimensions in this section assume a fuselage total width of 110–130mm (55–65mm half-width). The exact TPU start position must be recalculated and confirmed once the fuselage is built. See R21-4 (Commissioning Parameters) before cutting any spars.

Wing pair dimensions:

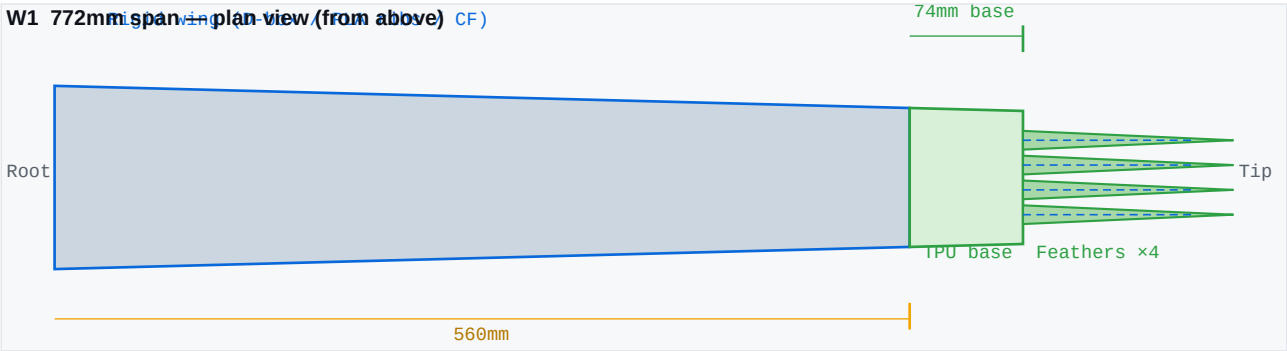
Pair	Span	TPU start (from root)	TPU total	Base (35%)	Feathers (65%)
W1	772mm	560mm	212mm	74mm	138mm
W2	749mm	560mm	189mm	66mm	123mm
W3	735mm	560mm	175mm	61mm	114mm
W4	712mm	560mm	152mm	53mm	99mm

The 35% / 65% base-to-feather split:

The base strip occupies the first 35% of the TPU section measured from the rigid wing boundary. It is solid, maintains the full S1223 cross-section profile, and structurally transitions from the last PLA rib to the feather zone. The carbon fibre trailing-edge rod terminates at the beginning of the base strip — it does not continue into the feather zone.

The remaining 65% is the feather zone — four cantilevered fingers with 4mm slots between them. These are the surfaces that perform the clap-and-fling and generate the aerodynamic slot-vortex benefit.

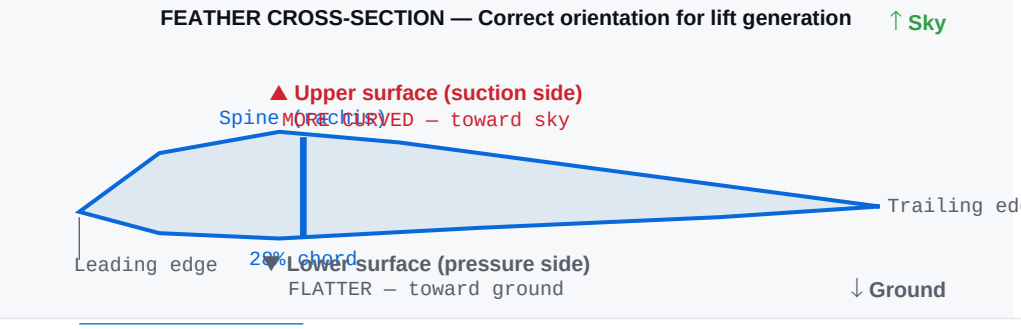
W1 plan view (representative — all pairs use same proportions):



Blue = rigid wing · Light green = TPU base · Dark green = feather fingers · Dashed = spine

R21-2 Feather Cross-Section — Orientation and Profile

■ **IMPORTANT** The feather cross-section has a MORE CURVED upper surface and a FLATTER lower surface. This is IDENTICAL to the standard aerofoil convention used on the rest of the wing. The upper surface (suction side) faces toward the sky and generates lift by low pressure. The lower surface (pressure side) faces toward the ground. This point is stated explicitly because the cross-section of a feather held in isolation — rachis pointing upward — can appear to have a curved underside. Once installed in the wing, curved side is UP.



The feather cross-section viewed from the root looking outward. The spine (rachis) is placed at 28% chord from the leading edge, creating the asymmetry that generates lift. Curved upper surface = suction side = sky. Flatter lower surface = pressure side = ground.

Why the spine offset creates the correct profile automatically:

The spine sits at 28% chord — the same position as a wing spar. The leading vane (LE to spine, 28% of chord) is short and relatively stiff. The trailing vane (spine to TE, 72% of chord) is long and more flexible. Under aerodynamic load during the downstroke, the trailing vane deflects slightly upward relative to the spine, generating positive camber: curved top, flatter bottom. The feather self-shapes into a lifting aerofoil under load. This is the same passive aeroelastic mechanism that real bird feathers use — no precise aerofoil engineering is required.

Cross-section specification:

Profile type	Asymmetric cambered section — NOT symmetric, NOT flat plate
Spine position	28% chord from the leading edge (same as wing spar quarter-chord rule)
Upper surface	MORE CURVED — suction side — faces sky — generates lift by low pressure
Lower surface	FLATTER — pressure side — faces ground — higher pressure
Leading edge	Rounded — radius approximately 15% of local chord. NOT sharp.
Trailing edge	Tapers to 0.5–1.0mm. Not knife-sharp — TPU at that thickness is already flexible and a knife edge wou
Max thickness	2.5mm at feather root (at spine location), tapering to ~0.8mm at 60% of feather length, then to ~0.3mm
Camber	~5% positive camber created naturally by spine offset under load
Aerofoil family	Simplified cambered plate — not full S1223. S1223 accuracy is not required at feather chord widths of

● **KEY POINT** The feather does NOT need to replicate the S1223 precisely. At chord widths of 8–12mm and the Reynolds numbers of the feather section (Re ~3,000–8,000), a simple cambered cross-section with a rounded nose performs comparably to an engineered aerofoil. The biological analogue (primary feather of a jackdaw) achieves lift comparable to manufactured aerofoils at these Re — Feather aerodynamics suggest importance of lift and flow predictability over drag minimisation (Scientific Reports, 2026).

R21-3 Material and Print Specification

One-piece design — single material throughout:

The entire TPU section — base strip and all four feathers — is printed as a single piece. There is no joint between base and feathers, no bonding, and no shore-hardness transition. A single Shore hardness throughout eliminates material interfaces that could delaminate under cyclic load.

Material	TPU (thermoplastic polyurethane) — filament grade
Shore hardness	85A (preferred) or 90A (acceptable)
Why 85A	85A gives feathers that deflect easily under clap loads (~120–180mN for 12mm sideways deflection). Th
Why not softer	70A–75A is too floppy — feathers oscillate in flight airflow at 1.87 Hz and develop parasitic flutter. 85A h
Why not harder	95A–98A feathers resist deflection (>400mN). Clap force becomes abrupt rather than gradual. Structural
Foamed TPU	Acceptable if your printer reliably produces foamed TPU. Foamed 90A behaves like solid 80A and saves
Recommended brand	Polymaker PolyFlex TPU95 (Shore 95A — use the 85A grade), Bambu TPU (85A grade), or Fiberlogy F
Mass per section	W1: ~12–15g W2: ~11–13g W3: ~10–12g W4: ~9–11g
Total (8 sections)	~88–104g — approximately 2–2.5% of MTOW

Print settings:

Print orientation	FLAT — span runs horizontally across the bed, chord vertical. Do NOT print standing up. Layer lines mu
Nozzle	0.4mm · Direct drive extruder strongly preferred
Layer height	0.15mm for surface quality on the aerofoil surfaces. 0.2mm acceptable for base strip only.
Perimeters	Base strip: 4 perimeters, 25% gyroid infill Feathers (inner 40% of length): 3 perimeters, 20% gyroid Feathers (outer 60% of length): 2 perimeters, 0% infill (hollow shell) Feather tips (last 20mm): 2 perimeters, 0% infill, minimum wall thickness 0.8mm
Print speed	25–35 mm/s for TPU. Slower is better for surface finish.
Temperature	Per filament datasheet. Typical: nozzle 220–235°C, bed 30–40°C.
Cooling	Minimum cooling during print — TPU needs time to bond between layers. Set part cooling fan to 20–30%
Retraction	Minimal — 0.5–1.0mm at 20–25 mm/s. More retraction causes stringing in TPU.
Supports	None required if printed flat on bed.

■ **CAUTION** Print all eight TPU sections (one per wing) before beginning attachment. Inspect each for delamination, particularly at the feather root junction where the change in infill density creates a stress concentration. Hold each feather tip and flex it 15mm sideways ten times. Any delamination shows as a white stress mark or audible crack — discard and reprint.

R21-4 Attachment to the Rigid Wing

The TPU section is a slip-over attachment — it slides onto the end of the CF spar and is secured with a single M2 bolt and a drop of CA. No permanent bonding to the rigid wing. Field replacement takes approximately 30 seconds.

1 **Prepare the spar end**

The CF leading-edge spar (2.0mm OD, 1.0mm ID tube) extends through the rigid wing to a point 5mm past the last PLA rib. Leave this 5mm extension — it is the socket for the TPU section. If the spar was previously cut flush to the last rib, the rib must be repositioned 5mm inboard before the TPU section can be attached.

2 **Print the socket into the TPU base**

The TPU section has a 2.2mm ID socket moulded into its leading-edge root face, running 15mm deep along the span axis. This accepts the 2.0mm CF spar with 0.1mm clearance each side. The socket is aligned with the leading-edge position of the S1223 profile at the TPU root. Similarly, a 1.6mm slot is moulded into the trailing-edge root face to accept the 1.5mm CF trailing-edge rod, which terminates 2mm proud of the last PLA rib. The slot runs 10mm deep.

3 **Dry fit**

Slide the TPU section onto the spar and rod without adhesive. Check that the upper surface (curved, suction side) faces the correct direction — toward the sky in the installed orientation. Check that the feather fingers are aligned spanwise (not rotated). The root face of the TPU section should seat flush against the face of the last PLA rib.

4 **Mark and drill M2 bolt hole**

With the TPU section in the correct dry-fit position, mark through the existing M2 hole in the last PLA rib attachment lug onto the TPU base strip. Remove the TPU section and drill a 2.2mm through-hole. The bolt runs chord-wise through the rib lug and the TPU base, perpendicular to the span.

5 **Apply adhesive and secure**

Apply one drop of thin CA to the inside of the CF spar socket in the TPU section. Slide the TPU section onto the spar. Seat the root face flush to the rib face. Insert and tighten the M2 bolt finger-tight plus one quarter turn. Do not overtighten — TPU compresses under bolt load and may crack the socket wall if over-torqued. Allow CA to cure 5 minutes before handling.

6 **Field replacement**

To replace a damaged TPU section: unscrew the M2 bolt, flex the TPU section slightly to break the CA bond at the socket, and pull off. Clean the spar end with isopropyl alcohol. Install new section as above. Carry one spare TPU section per wing pair in the field kit.

◆ **NOTE** The CF spar extends through the TPU base strip and is the primary structural connection. The M2 bolt prevents rotation and the CA bond prevents axial pull-out. Neither alone is sufficient — both are required.

R21-5 Commissioning Parameters — Required Before First Flight

■ **IMPORTANT** Two physical measurements must be entered into the flight software before TC-01 (bench check). The wing guard cannot be calibrated without them. The pre-flight check will fail with an explicit error message until both are confirmed. Do not bypass this check.

Parameter 1 — Fuselage total width:

Measure the distance between the port and starboard wing spar attachment points — outer face to outer face, at the attachment bracket level. This is the dimension that determines where along the wing span the port and starboard membranes first approach each other.

What to measure	Distance between outer faces of port and starboard spar attachment brackets at the fuselage wall, in mil
Expected range	100–140mm total width. Target: 110–120mm.
Measurement tool	Vernier calliper or digital calliper. Measure three times, use mean.
Software entry	GS: wc_set_fuselage_width [measured_mm] Example: wc_set_fuselage_width 118
Effect	Recalculates contact angles, rigid amplitude limits, and TPU clap limits for all four wing pairs. Printed co
When to measure	After fuselage is assembled and spar brackets are bolted in. Before wings are attached.

Parameter 2 — TPU section start position:

Measure the distance from the spar pivot point (centre of the root attachment bolt) to the innermost face of the TPU section when installed. This is the boundary between rigid wing structure (D-box, PLA ribs, CF rod) and the flexible TPU material.

What to measure	Distance from spar pivot (root bolt centre) to innermost TPU face, along the span axis, in millimetres
Expected range	W1: 555–565mm W2: 534–544mm W3: 520–530mm W4: 497–507mm (if all pairs built to same proportional specification)
Measurement tool	Steel rule or digital calliper. Measure with wing horizontal.
Software entry	GS: wc_set_tpu_start [measured_mm] Applies to all four pairs. If pairs differ, specify pair: GS: wc_set_tpu_start [mm] [pair_0-3] Example: wc_set_tpu_start 560 (all pairs) Example: wc_set_tpu_start 562 0 (W1 only)
Effect	Determines which contact zone is "rigid" (D-box / PLA ribs) vs "TPU" (membrane clap permitted). The wi
Critical check	The software will print a table showing the contact radius at 94°, 96°, 98°, 100° stroke. Confirm that 96° c GS: wc_geometry_report → check 'TPU zone' at 96°.
When to measure	After TPU sections are attached to all eight wings.

■ **CAUTION** If the 96° contact point falls in the rigid zone for any pair, the TPU section on that pair is too short — it must be reprinted starting 10–15mm further inboard. Do not attempt to fly with rigid-zone contact at commanded amplitudes. GS: wc_geometry_report shows the exact situation for each pair.

R21-6 Aerodynamic Performance of the TPU Section

Summary of research findings relevant to the MCO-1D feather design:

Slotted tip vs solid tip (lift coefficient)	+7.25% mean CL and +5.6% peak CL for a 20% gap configuration at low Reynolds number (Lynch et al., Bioinspir. Biomim., 2018). The MCO-1D feathers function as a slotted tip — this benefit applies.
Four-finger vs fewer (lift-to-drag)	Four-finger configuration outperforms one, two, or three fingers. L/D improvement up to 20% at low aspect ratio (AR 1.0–1.5) (ASCE Journal of Aerospace Engineering, 2024).
Slotted tip at flapping (flapping-specific)	As flapping frequency increases, slotted wingtip becomes MORE advantageous. The dynamically varying gaps during stroke cycle create additional unsteady aerodynamic benefit (Dan et al., Chinese Journal of Aeronautics, 2021).
Feather as standalone (individual feather)	An individual primary feather section generates lift comparable to manufactured aerofoils at the sa
Clap-and-fling (hover enhancement)	+15–25% lift in hover from clap-and-fling mechanism. TPU feathers performing the clap generate Flexible feathers outperform rigid surfaces for clap-and-fling because independent finger deflectio

What the pilot / operator experiences:

The aerodynamic effects of the TPU section are invisible during normal flight — the aircraft simply performs better at the tip than it would with a solid extension. The clap-and-fling benefit in hover is accessed through the ground station ("enable clap and fling") once the clap gate parameters have been commissioned. The wing guard monitors tip separation continuously and manages the amplitude envelope automatically.

◆ **NOTE** The slots between feathers are aerodynamically functional — they are not gaps to be minimised. Do not fill them with adhesive, sealant, or paint. Do not sand the feather surfaces smooth if the print surface is slightly rough — at low Reynolds numbers, minor surface texture has negligible effect and the print time saved by lower surface quality is not worth the risk of altering the intended geometry.

R21-7 Mass Budget and BOM Update

The TPU sections add to the wing group mass. Updated mass summary:

Item	Spec	Mass (est.)	Notes
TPU sections (8×)	Shore 85A, solid	+92–120g	W1×2, W2×2, W3×2, W4×2. Weigh each after printing.
M2 bolts (8×)	M2 × 8mm SS	~2g	One per section
CA adhesive	Thin, per section	~1g	Sockets only
Total addition	—	~95–123g	Approx 2.5% of 4.1kg MTOW — acceptable

BOM additions (items not in original BOM v18 or R20):

Item	Specification	Qty	Est. cost (USD)
TPU filament 85A	Polymaker PolyFlex TPU95 85A, 1.75mm, 0.5kg spool	1 spool	~$25–30
M2 × 8mm bolts	Stainless steel, pan head	1 pack (50)	~$5
M2 nuts	Stainless steel	1 pack (50)	~$4
Thin CA adhesive	Zap-a-Gap CA+ or equivalent, 1oz	1 bottle	~$6
Digital calliper	For commissioning measurement to ±0.1mm	1	Already in tool kit

R21-8 Cross-References and Document Dependencies

Wing Manufacturing Doc	Original document + Addendum R20: complete through STEP FINAL-2 before beginning TPU sectio
Commissioning Manual v21	After TPU sections are attached: enter wc_set_fuselage_width and wc_set_tpu_start before proceed
Software Reference v20	GS commands: wc_set_fuselage_width, wc_set_tpu_start, wc_geometry_report, clap_enable, clap_
Operator Manual v21	Section 11 (Wing Coordination and Clap-and-Fling) covers the pilot / operator interface for these sys
F-Register	F-306: TPU section commissioning parameters must be confirmed (see Commissioning Manual Add

MCO-1D Wing Manufacturing — Addendum R21

TPU Winglet Section — August 2026

This addendum supersedes any informal notes or earlier specifications for the wingtip region. All dimensions and proportions in this document assume a fuselage width in the 110–130mm range. The commissioning parameters in R21-5 are mandatory — the wing guard is calibrated from measured values, not estimates.